$$\therefore r_n = \frac{4\pi\varepsilon_0 (m\upsilon_n r_n)^2}{Zme^2}$$

$$r_n = \frac{4\pi\varepsilon_0 (n\hbar)^2}{Zme^2} = \frac{4\pi\varepsilon_0 n^2\hbar^2}{Zme^2}$$

$$r_n = \left(\frac{\varepsilon_0 h^2}{\pi me^2}\right)\frac{n^2}{Z} \qquad \left(\because \hbar = \frac{h}{2\pi}\right) \quad (9.16)$$

where $n \in \mathbb{N}$. Since, ε_0, h, e and π are constants. Therefore, the radius of the orbit becomes

$$r_n = a_0 \frac{n^2}{Z}$$

where $a_0 = \frac{\varepsilon_0 h^2}{\pi me^2} = 0.529\,\text{Å}$. This is known as Bohr radius which is the smallest radius of the orbit in hydrogen atom. Bohr radius is also used as unit of length called Bohr. 1 Bohr = 0.53 Å. For hydrogen atom (Z = 1), the radius of nth orbit is

$$r_n = a_0 n^2$$

For n = 1 (first orbit or ground state),

$$r_1 = a_0 = 0.529\ \text{Å}$$

For n = 2 (second orbit or first excited state),

$$r_2 = 4a_0 = 2.116\ \text{Å}$$

For n = 3 (third orbit or second excited state),

$$r_3 = 9a_0 = 4.761\ \text{Å}$$

and so on.

Thus the radius of the orbit from centre increases with n, that is, $r_n \propto n^2$ as shown in Figure 9.18.

Further, Bohr's angular momentum quantization condition leads to

$$\frac{m\upsilon_n a_0 n^2}{Z} = n\frac{h}{2\pi} \quad \left[\because r_n = a_0\frac{n^2}{Z}\right]$$

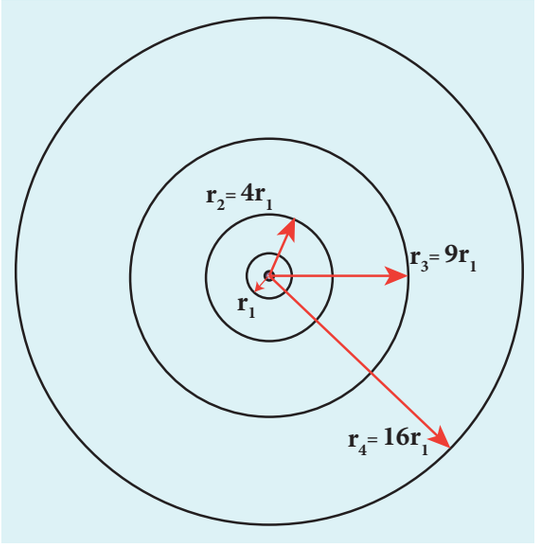


Figure 9.18 Variation of radius of the orbit with principal quantum number

$$\upsilon_n = \frac{h}{2\pi m a_0}\frac{Z}{n}$$

in atomic physics $\upsilon_n \propto \frac{1}{n}$

Note that the velocity of electron decreases as the principal quantum number (orbit number) increases as shown in Figure 9.19. This curve is the rectangular hyperbola. This implies that the velocity of electron in ground state is maximum when compared to that in excited states.

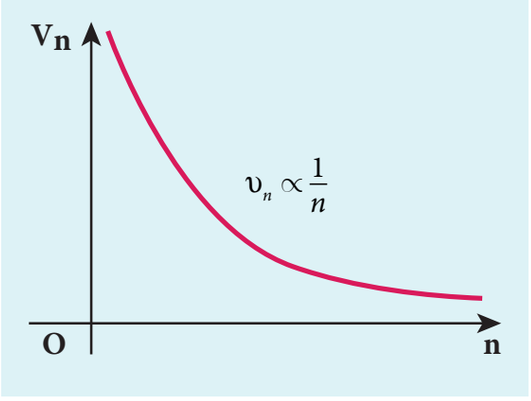


Figure 9.19 Variation of velocity of the electron in the orbit with principal quantum number

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